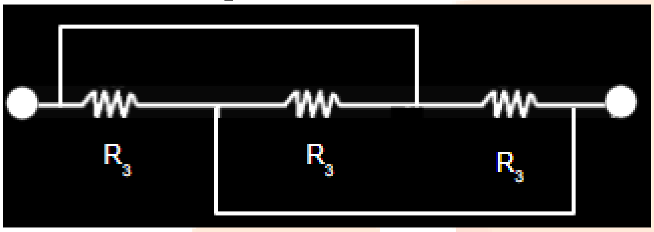
JEE-Main-28-01-2025 (Memory Based)
[MORNING SHIFT]

Physics

Question: The equivalent resistance across A and B is



Options:
(a) R/3
(b) R/2
(c) R/9
(d) R
Answer: (c)

Question: A uniform wire of linear charge density λ is placed along y-axis. The locus of equipotential
Options:
(a) $x^2+y^2 + z^2$ = constant
(b) $x^2 + z^2$ = constant
(c) xyz = constant
(d) xy + yz + zx = constant
Answer: (b)

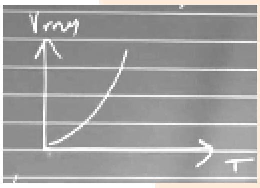
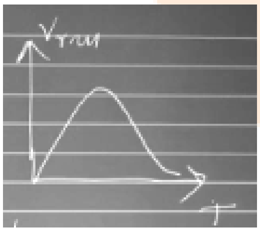
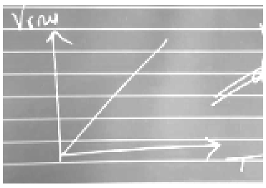
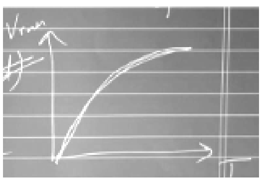
Question: Three carnot engine operating at different temperature ranging from
273 K → 473 K, $373 \xrightarrow{K} 473k$, 273 K → 373 K, Find % efficiency nearly
Options:
(a) 42, 21, 26
(b) 52 , 24 , 28
(c) 22, 21 , 26
(d) 50, 21, 26
Answer: (a)

Question: Which of the following reactions is correct? (Where symbols have their usual meanings)
Options:
(a) n → p+e^- +v

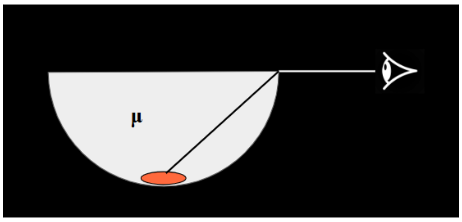
(b) n → p+e^{+} +v
(c) $n \rightarrow p + e^{+} + \overline{v}$
(d) $n \rightarrow p + e^{-} + \overline{v}$
Answer: (d)

Question: Two disc having equal mass, radius of one is twice the other find the ratio of I_1/I_2 ($r_2 = r_1$ x 2)
Options:
(a) 1/4
(b) 4/1
(c) 2/1
(d) 1/2
Answer: (a)

Question: An ice water mixture is present at 273 Kelvin at the initial pressure equal to atmospheric pressure if the pressure is doubled keeping the temperature constant then
Options:
(a) More ice will melt
(b) More ice water will convert to ice
(c) Water will vapourize completely
(d) Water will completely convert to ice
Answer: (a)

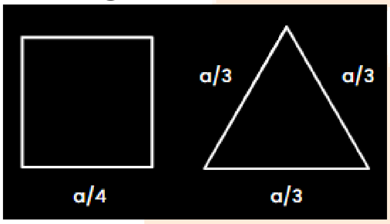
Question: The graph of root mean square velocity versus temperature is ?
Options:
(a) 
(b) 
(c) 
(d) 
Answer: (d)

Question: A coin is placed at the bottom of a hemispherical container filled with a liquid of refractive index u. Find the least refractive index if the coin is visible to an observer at E.



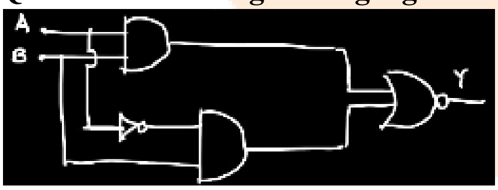
Options:
(a) √3
(b) √2
(c) √3/2
(d) 2√3
Answer: (b)

Question: In the given figure, the square and the triangle have the same resistance per unit length. Find the ratio of their resistances about adjacent corners.



Options:
(a) 32/27
(b) 27/32
(c) 8/9
(d) 9/8
Answer: (b)

Question: For the given logic gates combination correct truth table will be



Options:
(a)

A	B	Y
0 0 1 1	0 1 0 1	0 0 1 1

(b)

A	B	Y
0 0 1 1	0 1 0 1	0 1 0 1

(c)

A	B	Y
0 0 1 1	0 1 0 1	0 1 1 1

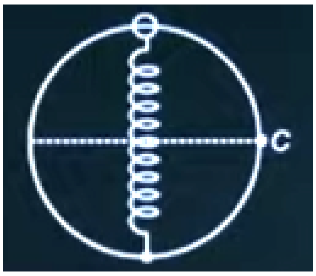
(d)

A	B	Y
0 0 1 1	0 1 0 1	1 0 1 0

Answer: (d)

Question: Assertion : Work done by central force is independent of path.
Reason : Potential energy is associated with every force.
Options:
(a) Both Assertion and Reason are correct
(b) Assertion is correct, Reason is incorrect
(c) Assertion is incorrect, Reason is correct
(d) Both Assertion and Reason are incorrect
Answer: (d)

Question: There is a smooth ring of radius R in the vertical plane. A spring of natural length R and elastic constant K is vertical across along a diameter. The free end is connected to bead of mass m and when slightly disturbed it reaches point C with speed where V is

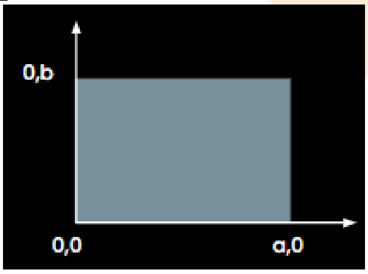


Options:

(a) $\sqrt{\dfrac{KR^2\left(\sqrt{2}-1\right)+2mgR}{m}}$
(b) $\sqrt{\dfrac{2KR^2\left(\sqrt{2}-1\right)+2mgR}{m}}$
(c) $\sqrt{\dfrac{2KR^2\left(\sqrt{2}-1\right)+mgR}{m}}$
(d) $\sqrt{\dfrac{KR^2\left(\sqrt{2}-1\right)+mgR}{m}}$
Answer: (b)

Question: A proton moving at kinetic energy E has the same kinetic energy as that of a photon. The ratio of de-Broglie wavelength of proton to photon is
Options:
(a) $\dfrac{1}{C}\sqrt{\dfrac{2E}{m}}$
(b) $\dfrac{1}{C}\sqrt{\dfrac{E}{2m}}$
(c) $\dfrac{2}{C}\sqrt{\dfrac{E}{m}}$
(d) $\dfrac{2}{C}\sqrt{\dfrac{2E}{m}}$
Answer: (b)

Question: Surface mass density varies as $\sigma=\dfrac{\sigma_0 x}{ab}$ **for the given plane sheet. Find the position of centre of mass for the distribution given**



Options:
(a) 2a/3, 2b/3
(b) 2a/3, b/2
(c) a/3, b/2
(d) a/2, b/2
Answer: (b)

Question: Identify the correct statement/s from the following list and select the appropriate option.
I) The coefficient of viscosity decreases as the temperature increases.
II) Terminal velocity of a spherical body falling through a viscous fluid depends on it radius r, it's density and acceleration due to gravity.
III) Terminal velocity of a spherical body falling through a viscous fluid depends only on density of body and density of fluid.
IV) The coefficient of viscosity is independent of temperature.
V) Viscous force acting on a spherical body is directly proportional to speed of body
Options:
(a) I, II , V
(b) III , V
(c) I , II , V
(d) III , VI , V
Answer: (c)

Question: When light of 600 nm is used, the 10th bright fringe is 10 mm from central bright. Find the distance from central max if light of 660 nm is used.
Options:
(a) 10 mm
(b) 11 mm
(c) 12 mm
(d) 14 mm
Answer: (b)

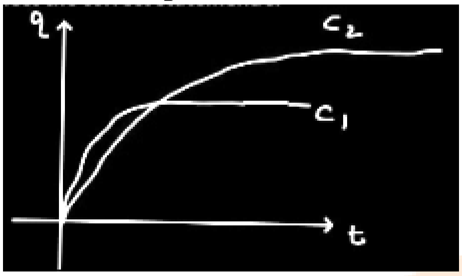
Question: Dispersion without deviation is produced by two thin (small angled) prisms which are combined together. One prism has angle 4° and refractive index 1.56. If the other prism has refractive index 1.7, what is it's angle ?
Options:
(a) 3.2°
(b) 2.2°
(c) 5.5°
(d) 8°
Answer: (a)

Question: 3 infinite wires each having linear charge density λ are kept, one each at x, y and z axis respectively. Then the locus of equipotential surface will be
Options:
(a) xyz = constant
(b) xy + yz + zx = constant
(c) (x+y) (y+z)(z +x) = constant
(d) $(x^2+y^2)(y^2+z^2)(z^2+x^2)$ = constant
Answer: (d)

Question: If the area of 2.5 mm x 5 mm sheet is calculated using a screw gauge of pitch 0.75 mm and 15 division on circular scale, then the error in the calculation of area is x/100 . Find x after rounding off.
Options:
(a) 38
(b) 48

(c) 58
(d) 28
Answer: (a)

Question: If two capacitors C_1 and C_2 are in parallel combination. Variation of charge stored on capacitor with time is show in the graph. Select the correct statement/s.



(1) $C_1 > C_2$
(1) $C_1 < C_2$
(3) $U_1 > U_2$
(4) $U_1 < U_2$
Options:
(a) 1 & 3
(b) 1 & 4
(c) 2 & 3
(d) 2 & 4
Answer: (d)

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[MORNING SHIFT]
Chemistry

Question: The product A and B in the following reactions, respectively

$(A) \xleftarrow{AgNO_2} CH_3CH_2CH_2Br \xrightarrow{AgCN} B$

Options:

(a) CH_3 - CH_2 - CH_2 - ONO, CH_3 - CH_2 - CH_2 - CN
(b) CH_3 - CH_2 - CH_2 - NO_2, CH_3 - CH_2 - CH_2 - NC
(c) CH_3 - CH_2 $\longrightarrow$ + CH_2 - NO_2, CH_3 - CH_2 - CH_2CN
(d) CH_3 - CH_2 - CH_2 - ONO, CH_3 - CH_2 - CH_2 - NC

Answer: (b)

Question: The molecules having square pyramidal geometry are

Options:

(a) SbF_3 & PCl_5
(b) BrF_5 & $XeOF_4$
(c) BrF_5 & PCl_5
(d) SbF_5 & XeF_4

Answer: (b)

Question: The incorrect decreasing order of atomic radii is,

Options:

(a) Si > P > Cl > F
(b) Mg > Al > C > O
(c) Al > B > N > F
(d) Be > Mg > Al > Si

Answer: (d)

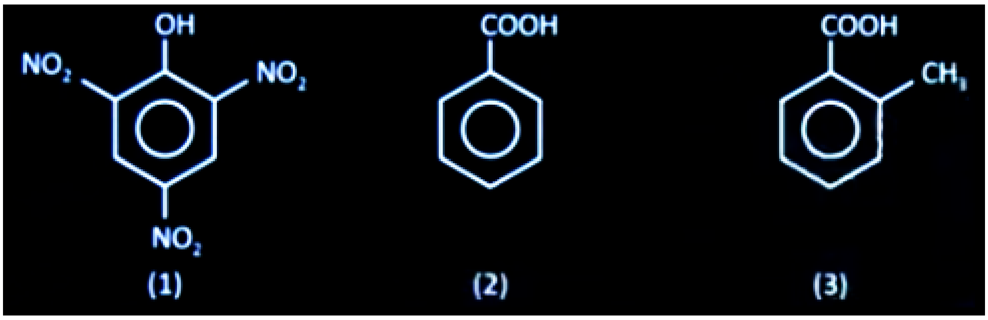
Question: Consider the following element in ln TI, Al, and Pb. The most stable oxidation states of elements with highest and lowest first ionisation enthalpies, respectively are

Options:

(a) +4 and +1
(b) +2 and +3
(c) +4 and +3
(d) +1 and +4

Answer: (d)

Question: What is the rate of reaction $CO_2(g)$ with aq. $NaHCO_3$ among the following?



Options:

(a) (1) > (2) > (3)

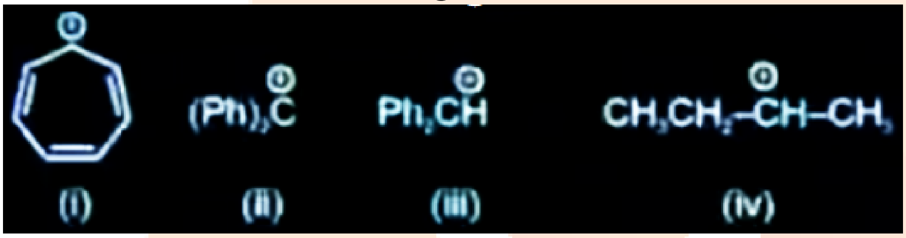
(b) (3) > (2) > (1)

(c) (1) > (3) > (2)

(d) (2) > (3) > (1)

Answer: (c)

Question: Consider the following carbocations



The correct increasing order of stability of these carbocations is:

Options:

(a) i < ii < iii < iv

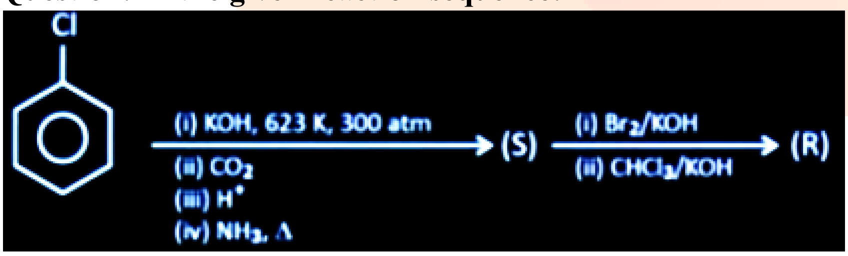
(b) iv < iii < ii < i

(c) ii < iii < iv < i

(d) iv < iii < i < ii

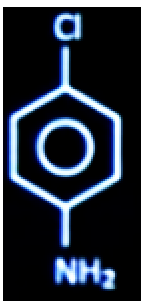
Answer: (b)

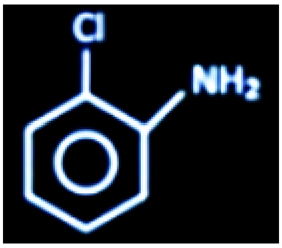
Question: In the given reaction sequence:

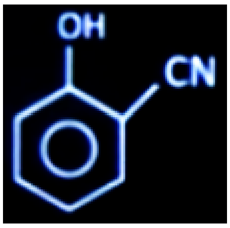


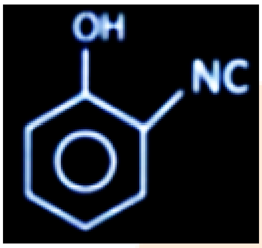
What is (R)

Options:

(a) 

(b) 

(c) 

(d) 

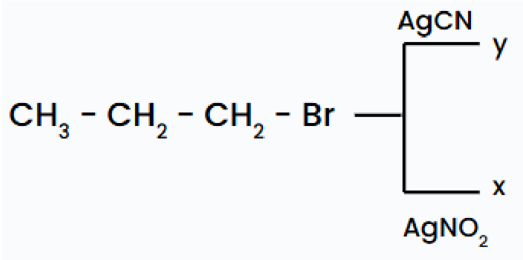
Answer: (d)

Question: Which of the following set of quantum numbers have same energy?
(a) n = 2, I = 2, m = + 1
(b) n = 2, I = 1, m = - 1
(c) n = 3, I = 2, m = 0
(d) n = 3, I = 2, m = 1
Options:
(a) a, b
(b) b, c
(c) c, d
(d) a, c
Answer: (c)

Question: Which has the same no. of unpaired e^- as no of lone pairs in ClF_3?
Options:
(a) V^{2+} , Ni^{2+}
(b) V^{3+}, Cu^{2+}
(c) Cu^{2+}, Ni^{2+}
(d) Ni^{2+}, V^{3+}
Answer: (d)

Question: Which will give a positive test in both acetone and acetaldehyde?
Options:
(a) 2, 4 DNP
(b) Tollen’s Reagent
(c) Fehling's solution
(d) Schiff's base test
Answer: (a)

Question: Consider the following reaction



The major product x and y respectively are

Options:

(a) $CH_3CH_2CH_2ONO$ & $CH_3CH_2CH_2CN$
(b) $CH_3CH_2CH_2NO_2$ & $CH_3CH_2CH_2CN$
(c) $CH_3CH_2CH_2NO_2$ & $CH_3CH_2CH_2NC$
(d) $CH_3CH_2CH_2ONO$ & $CH_3CH_2CH_2CN$

Answer: (c)

Question: Match the following column and choose the correct option.

	Column-I		Column-II
A	$H_2O_2 \rightarrow H_2O + O_2$	P	Combustion reaction
B	NaH 2♠ Na + H_2	Q	Disproportionation
C	$CH_4 + O_2 \rightarrow CO_2 + H_2O_2$	R	Decomposition reaction
D	$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$	S	Displacement reaction

Options:

(a) A-Q, B-P, C-R, D-S
(b) A-R, B-Q, C-S, D-P
(c) A-Q, B-R, C-P, D-S
(d) A-R, B-Q, C-P, D-S

Answer: (c)

Question: **Mass % of C, H, Cl in an organic compound is given below. Find its empirical mass**

Cl = 65 %
H = 1.8 %
C = 32.8%

Options:

(a) $C_3H_2Cl_2$
(b) C_3H_2Cl
(c) C_3HCl_2
(d) CH_2Cl_2

Answer: (a)

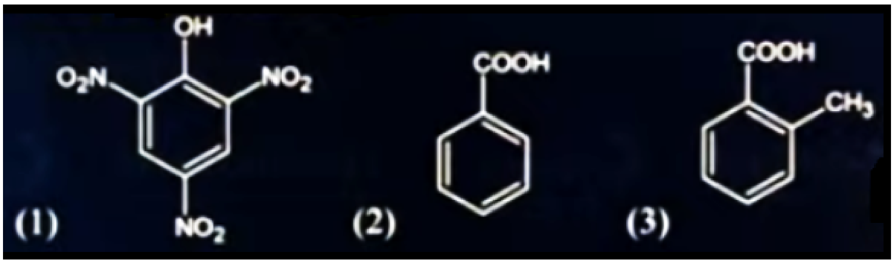
Question: A weak acid HA has degree of dissociation *x*. Which options gives the correct expression of ($pH - pK_a$)
Options:
(a) 0
(b) ($\log(1 + 2x)$)
(c) $log\left(\frac{x}{1-x}\right)$
(d) $log\left(\frac{1-x}{x}\right)$
Answer: (c)

Question: Both acetaldehyde and acetone (individually) undergo which of the following reactions,
(A) iodoform Reaction
(B) Cannizzaro Reaction
(C) Aldol condensation.
(D) Tollen's test
(E) Clemmensen Reduction
Options:
(a) A, C & E only
(b) A, D & E only
(c) A, B, C, D & E
(d) A & C only
Answer: (a)

Question: What is the freezing point of depression constant of a solvent 50g of which contain 1g of non-volatile solute (M.W: 256g/mol) and depression in freezing point is 0.4K
Options:
(a) 0.372K Kg mol^{-1}
(b) 4.213 K kg mol^{-1}
(c) 1.86K Kg mol^{-1}
(d) 5.12K Kg mol^{-1}
Answer: (d)

Question: Ice and water are placed in a closed container at a pressure at 1 atm and temperature 273.15K.
If the pressure of the container increases 2 times and the temperature is kept constant, than identify the correct observation from the following
Options:
(a) The amount of ice decreases
(b) Volume of system increases
(c) Liquid phase disappear completely
(d) Solid phase (ice) disappear completely
Answer: (d)

Question: What is the rate of reaction for releasing CO_2(g) with sq. $NaHCO_3$ among following?



Options:
(a) (1) > (2) > (3)
(b) (3) > (2) > (1)
(c) (1) > (3) > (2)
(d) (2) > (3) > (1)
Answer: (c)

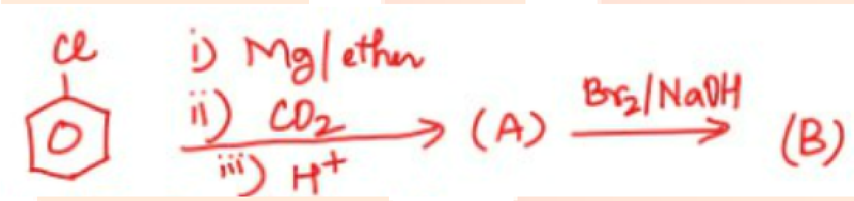
Question: 70% by mass solution of HNO_3 is taken having density 1.41 gm/ml. Calculate molarity (Rounded of to nearest integer)

Answer: (16)

Question: $\Delta_f H$ of H(g) is 218 kJ/mol, $\Delta_f H$ of O(g) is 249.2kJ/mol, $\Delta_f H$ of H_2O is -241.8 kJ/mol. What is the value of Bond Energy of O - H bond in H_2O in kJ/mol?

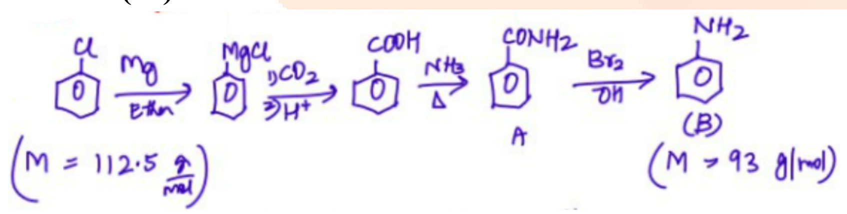
Answer: (463.5)

Question:



11.25 mg iv) NH_3/Δ
The mass of B is $x \times 10^{-1}$ mg. Find x.

Answer: (93)



112.5g chlorobenzene $\longrightarrow$ 93 g aniline
11.25 mg chlorobenzene $\longrightarrow \frac{93g}{10^{11.25}g} \times 11.25$
$= 9.3$ mg $= 93 \times 10^{-1}$ mg

JEE-Main-28-01-2025 (Memory Based)
[MORNING SHIFT]
Maths

Question: If $\int_{-\pi/2}^{\frac{\pi}{2}} \frac{96(x^2.cosx)}{1+e^x} dx$ **= a**π^3 **+**β **(where** $\alpha+\beta$ **are positive integers), then** $\alpha+\beta$ **equal to**

Options:
(a) 144
(b) 100
(c) 64
(d) 196
Answer: (b)

$$I = 96 \int_{-\pi/2}^{\pi/2} \frac{x^2 + \cos x}{1+e^x} dx$$

$$I = 96 \int_{-\pi/2}^{\pi/2} \frac{x^2 + \cos x}{1+e^{-x}} dx$$

$$2I = 96 \int_{-\pi/2}^{\pi/2} \left(x^2 + \cos x\right) dx$$

$$I = 96\left(\frac{x^3}{3} + \sin x\right)_0^{\pi/2} = 96\left(\frac{\pi 3}{24} + 1\right)$$

$$= 4\pi^3 + 96$$

$$\alpha + \beta = 100$$

Question: Number of ways to form 5 digit numbers greater than 50000 with the use of digits 0,1,2,3,4,5,6,7 such that sum of first and last digit is not more than 8, equal to
Options:
(a) 5119
(b) 5120
(c) 4607
(d) 4068
Answer: (c)

Available digit 0, 1, 2, 3, 4, 5, 6, 7

Middle three spaces can be filled in

1st place 5, then last place can be 0, 1, 2, 3 as sum is less than or equal to 8

So total no's = 1 × 512 × 4 = 2048

1st place 6, then last place can be 0, 1, 2

So total no. of ways = 1 × 512 × 3 = 1536

1st place 7, then last place can be 0, 1

So total no. of ways = 1 × 512 × 2 = 1024

So total no. of ways = 2048 + 1536 + 1024 = 4608

Excluding 50000, 4608 - 1 = 4607

Question: If $f(x)=\frac{2^x}{2^x+\sqrt{2}}, x \in R,$ then $\sum_{k=1}^{81} f\left(\frac{k}{82}\right)$ **is equal to**

Options:

(a) $81\sqrt{2}$

(b) 82

(c) $\frac{81}{2}$

(d) 42

Answer: (c)

$$f(x)=\frac{2^x}{2^x+\sqrt{2}} \Rightarrow f(x)+f(1-x)=1$$

$$s=\sum_{n=1}^{81} f\left(\frac{k}{82}\right)$$

$$s=\sum_{n=1}^{81} f\left(\frac{82-x}{82}\right)=\sum_{n=1}^{81} f\left(1-\frac{4}{82}\right)$$

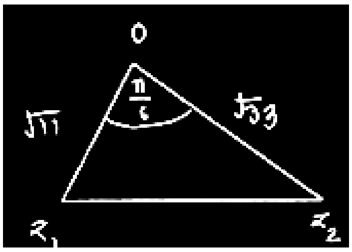
$$2s=\sum_{n=1}^{81} f\left(\frac{k}{82}\right)+f\left(1-\frac{k}{82}\right)$$

$$=\sum_{n=1}^{81} 1=81$$

$$s=\frac{81}{2}$$

Question: $z_1 = \sqrt{3} + 2\sqrt{2}i$ & $\sqrt{3}\left|Z_1\right| = \left|Z_2\right|$ **and arg(z_2) =arg(z_1) +** $\frac{\pi}{6}$ **then area, of triangle with vertices z_1, z_2 and origin**

Solution:



$$\text{Area} = \frac{1}{2}\sqrt{11}\,.\,\sqrt{33}\,\sin\frac{\pi}{6}$$

$$= \frac{1}{2}\,.11\sqrt{3}.\,\frac{1}{2}$$

$$= \frac{11\sqrt{3}}{4}$$

Question: The relation R={(x,y)| x, y $\in$ z, x + y = even} then R is

Options:

(a) Equivalence

(b) Reflexive & Transitive but - not Symmetric

(c) Symmetric & Transitive but not reflexive

(d) Reflexive & symmetric but not transitive

Answer: (a)

$$R = \{(x,y) : x + y = \text{even}, x, y \in z\}$$

$$\text{Reflexive as } x + x = \text{even}$$

$$\text{symmetric as } x + y = y + x$$

$$\text{Transitive as } x + y = \text{even } y + z = \text{even}$$

$$x + 2y + z = \text{even}$$

$$x + z = \text{even}$$

$$Equivalence.$$

Question: $\cos\left(\sin^{-1}\frac{3}{5}+\sin^{-1}\frac{5}{13}+\sin^{-1}\frac{33}{65}\right)$ **is equal to**

Options:

(a) 0

(b) 1

(c) $\frac{32}{65}$

(d) $\frac{33}{65}$

Answer: (a)

$$\cos\left(\sin^{-1}\frac{3}{5} + \sin^{-1}\frac{5}{13} + \sin^{-1}\frac{33}{65}\right)$$

$$\cos\left(\tan^{-1}\frac{3}{4} + \tan^{-1}\frac{5}{12} + \sin^{-1}\frac{33}{65}\right)$$

$$\cos\left(\tan^{-1}\frac{\frac{3}{4}+\frac{5}{12}}{1-\frac{3}{4}\times\frac{5}{12}} + \sin^{-1}\frac{33}{65}\right)$$

$$\cos\left(\tan^{-1}\frac{14\times 4}{33} + \cot^{-1}\frac{56}{33}\right)$$

$$\cos\frac{\pi}{2} = 0$$

Question: $\int_{0}^{x} tf(t)dt = x^2 f(x) f(2)3, f(6) = ?$

Solution :

$$\int_{0}^{x} tf(x)dt = x^2 f(x)$$

$$xf(x) = 2xf(x) + x^2 f'(x)$$

$$\Rightarrow xf'(x) = -f(x)$$

$$\Rightarrow \frac{f'(x)}{f(x)} = \frac{-1}{x}$$

$$\Rightarrow \ln f(x) + \ln x = \ln c$$

$$\Rightarrow xf(x) = c$$

$$2f(2) = 6f(6)$$

$$\Rightarrow f(6) = 1$$

Question: Area of region {(x, y) : $0 \le y \le 2|x| + 1$, $0 \le y \le x^2 + 1$, $|x| \le 3$}

Options:

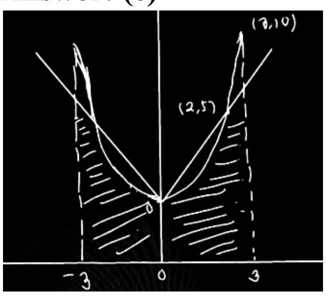
(a) $\frac{17}{3}$

(b) $\frac{32}{3}$

(c) $\frac{64}{3}$

(d) $\frac{80}{3}$

Answer: (c)



$$A = 2\left[\int_0^2 (x^2+1)dx + \int_2^3 (2x+1)dx\right]$$

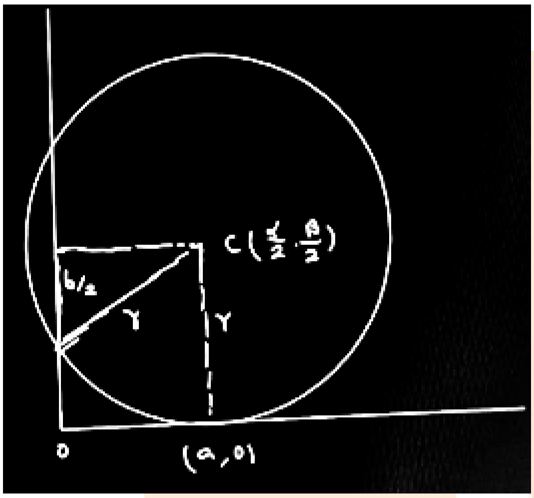
$$= 2\left[\left(\frac{x^3}{3}+x\right)_0^2 + \left(x^2+x\right)_2^3\right]$$

$$= 2\left(\frac{8}{3}+2+9+3-4-2\right) = 2\left(8+\frac{8}{3}\right)$$

$$= 2 \times \frac{32}{3} = \frac{64}{3}$$

Question: $x^2 - \alpha x + \beta y - \gamma + y^2 = 0$ if touches x axis at a point A(a, 0)& cut intercept of bv. find value of a and b on y axis

Solution :



$$r = \sqrt{\frac{\alpha^2}{4} + \frac{\beta^2}{4} - \gamma} = \frac{\beta}{2}$$

$$\frac{\alpha^2}{4} - \gamma = 0$$

$$\alpha^2 = 4\gamma, \frac{\alpha}{2} = a$$

$$\alpha = 2a$$

$$2\sqrt{\frac{13^2}{4} - \gamma} = b$$

$$\frac{\beta^2}{4} = \gamma = \frac{b^2}{4}$$

$$b^2 = \beta^2 - 4\gamma$$

$$(2a, b^2) = (\alpha, \beta^2 - 4\gamma)$$

Question: $x^2 + |2x - 3| - 4 = 0$ has sum of squares of roots

Solution :

$x^2 + |2x - 3| - 4 = 0$

If $x \geq \frac{3}{2}, x^2 + 2x - 7 = 0$

$x = \frac{-2 \pm \sqrt{4+28}}{2} = \frac{-24\sqrt{2}}{2}$

$= -1 \pm 2\sqrt{2}$

$x = 2\sqrt{2} - 1$

If $x < \frac{3}{2}, x^2 - 2x - 1 = 0$

$x = \frac{2 \pm 2\sqrt{2}}{2} = 1 + \sqrt{2}$

$x = 1 - \sqrt{2}$

$\text{sum of squares of roots} = \left(2\sqrt{2} - 1\right)^2 + \left(1 - \sqrt{2}\right)^2$

$= 12 - 6\sqrt{2}$

Question: The image of point (4, 4, 3) in the plane $= \frac{x-1}{2} = \frac{y-2}{1} = \frac{z-1}{3}$ **is (α, β, γ) find $\alpha + \beta + \gamma$.**

Solution :

Given line is x = 1 + 2t, y = 2 + t, z = 1 + 3t

Vector joining (4, 4, 3) to the general point on line

(1 + 2t, 2 + t, 1 + 3t) is $\vec{v}$ (- 3 + 2t, - 2 + t, - 2 + 3t)

Orthologous projection $\vec{v}.\vec{d} = 0$ d is direction

Vector of the line as $\vec{d} = 2i + j + 3k$

So (-3 + 2t)2 + (-2 + t) $\cdot$ 1 + (-2 + 3t) $\cdot$ 3 = 0

-14 + 14t = 0 ⇒ t = 1

Point of projection is (3, 3, 4)

Vector joining the projection point (3, 3 4) and (4, 4, 3) - (1, 1, 1)

So image point = (3, 3, 4) + (1, 1, -1) = (2, 2, 5)

Sum of coordinates = 2 + 2 + 5 = 9

Question: $a_0 = 1$, $a_1 = \frac{1}{2}$, $2a_n = 5a_{n-1} - 3a_{n-2}$ find $\sum_{i=1}^{100} a_n$.

Options:

(a) $3a_{100} + 197$

(b) $2a_{100}$ - 1000
(c) $3a_{99}$ - 100
(d) $2a_{99}$ - 100
Answer: (a)

$$a_0 = 1, a_1 = \frac{1}{2}, 2a_n - 3a_{n-1} = 2a_{n-1} - 3a_{n-2}$$

$$\Rightarrow 2a_n - 3a_{n-1} = 2a_1 - 3a_0$$

$$= 1 - 3 = -2$$

$$2\sum_{n=1}^{100} a_n - 3\sum_{n=1}^{100} a_{n-1} = -200$$

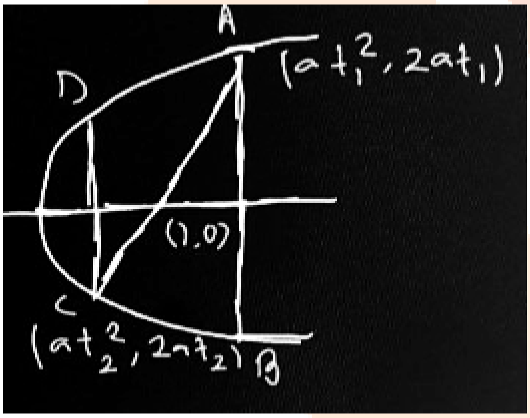
$$2s = 3[a_0 + s - a_{100}] = -200$$

$$2s = 3s - 3 + 3a_{100} = -200$$

$$s = 3a_{100} + 197$$

Question: Vertices of a trapezium lies on a parabola y2 = 4x one of its diagonal passes through (1, 0), and its length is $\frac{25}{4}$. Parallel sides are parallel to y. Find its area of trapezium.

Solution :



$$t_1 t_2 = -1$$

$$\sqrt{\left(t_1^2 + t_2\right)^2 + \left(2t_1 - 2t_2\right)^2} = \frac{25}{4}$$

$$|t_1 - t_2|\sqrt{\left(t_1 + t_2\right)^2 - 4t_1 t_2} = \frac{25}{4}$$

$$\left(t_1 - t_2\right)^2 = \frac{25}{4}$$

$$t_1 - t_2 = \frac{5}{2}$$

$$r_1 + \frac{1}{t_1} = \frac{5}{2}$$

$$t_1 = 2, t_2 = -\frac{1}{2}$$

$$A = \frac{1}{2}(8 + 2) \times \left(4 - \frac{1}{4}\right) = \frac{1}{2} \times 10 \times \frac{15}{4} = \frac{75}{4}$$

Question: A bag has 7 good and 3 defected oranges. 2 oranges are chosen randomly. Find the variance of oranges being defected.
Solution:

7 Good, 3 Defective

x	f
0	${}^7C_2/{}^{10}C_2 = \frac{7}{15}$
1	${}^3C_1 \cdot {}^7C_1/{}^{10}C_2 = \frac{7}{15}$
2	${}^3C_2/{}^{10}C_2 = \frac{1}{15}$

$$\bar{x} = 0 + \frac{7}{15} + \frac{2}{15} = \frac{9}{15} = \frac{3}{5}$$

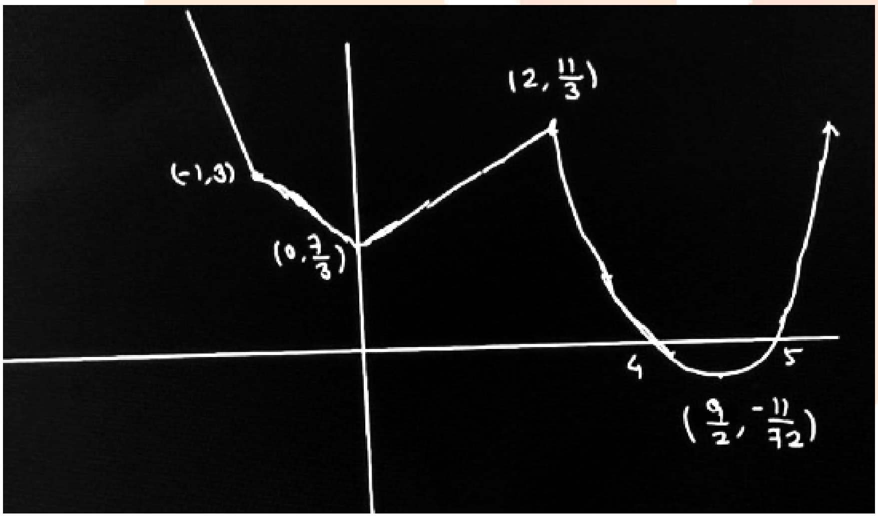
$$Var = \left(0 + \frac{7}{15} + \frac{4}{15}\right) - \left(\frac{3}{5}\right)^2 = \frac{11}{15} - \frac{9}{25}$$

$$= \frac{110-54}{150} = \frac{56}{150} = \frac{28}{75}$$

Question: The sum of all local minimum values of the function

$$\begin{cases} 1-2x & x<-1 \\ \frac{1}{3}(7+2|x|) & -1 \le x \le 2 \\ \frac{11}{18}(x-4)(x-5) & x>2 \end{cases}$$ is

Solution :



$$\text{Sum of local minimum values} = \frac{7}{3} - \frac{11}{72}$$

$$= \frac{168-11}{72} = \frac{157}{72}$$